From raw thermal logs to engineering insight

A practical, local-first workspace for exploring, comparing and reporting thermal test runs.

IMPORT → VISUALIZE → ANALYZE → COMPARE → REPORT

Windows engineering pilot • Browser interface • No separate Python install for test users

The engineering problem

Scattered data

Logger exports vary by file format, channel and duration.

Slow investigation

Finding transitions, stable periods and anomalies takes time.

Hard to compare

Different test windows make run-to-run review difficult.

Manual reporting

Charts and calculations often need to be recreated.

Traceability gaps

Reviewers need settings, inputs and caveats alongside results.

Pilot opportunity

Bring a repeatable workflow into one local tool.

THERMAL TEST INTELLIGENCE CENTER

What TTIC does — in one view

01 Test Setup

Load CSV/XLSX, inspect channels and timestamps.

02 Visualization

Interactive multi-channel plots with dual axes.

03 Engineering Analytics

Statistics, heating/cooling rates, stability screening.

04 Event Intelligence

Command-step detection and response timing.

05 Reports & Export

Engineering-ready Excel/PDF outputs.

06 Test Comparison

Compare two runs using flexible time windows.

The workflow: six steps, one workspace

1 Load

Import logger files

2 Explore

Select and plot channels

3 Measure

Compute raw-data metrics

4 Investigate

Screen steps and stability

5 Compare

Align the analysis scope

6 Share

Export results and settings

Data in: flexible, but inspectable

File formats

CSV and Excel logger exports.

Channel mapping

Choose relevant measured and command channels.

Time handling

Parse elapsed time and select analysis windows.

Quality review

Check missing values, timestamps and channel suitability.

Local processing

Core workflow operates on the user’s machine.

Engineering context

Interpret units and test conditions before comparing.

Visual investigation without rebuilding charts

Interactive traces

Choose channels, colors and dual-axis views.

Focused windows

Zoom into start-up, transitions or steady regions.

Display smoothing

Optional visual aid; calculations use raw selected data.

Run overlays

See two test profiles together.

Independent windows

Compare chosen periods even when runs differ in length.

Exportable views

Carry the evidence into review reports.

Engineering analytics: what is measured

Descriptive statistics

Min, max, mean, spread and sample count.

Heating / cooling

Overall and interval-based positive/negative rates.

Stability screening

Low-variation windows and grouped periods.

Response timing

t10, t50, t90 and 10–90% transition time.

Configurable settings

Windows, thresholds and channel choices.

Important boundary

Screening ≠ equilibrium proof; timing ≠ pass/fail.

Event intelligence: from command to response

COMMAND STEP → BASELINE → FINAL REFERENCE → t10 / t50 / t90

Find the event

Detect command steps or specify event time manually.

Measure response

Use selected response channel and raw threshold crossings.

Review assumptions

Check baseline, reference tail, observation window and signal noise.

Compare tests fairly — not just visually

Full Test

Review each run across its complete duration.

Common Duration

Limit both to a shared observation length.

Independent Windows

Choose different relevant windows per test.

Separate calculations

Statistics are recomputed for each selected scope.

Avoid false precision

Do not infer equivalence from an overlay alone.

Interpret with context

Match units, setup, loading and environment.

Reports that engineers can review

Excel workbook

Executive summary, metrics, supporting tables, chart data and settings.

PDF summary

Compact comparison narrative, tables and visual evidence.

Response detail

Include t10, t50, t90 and 10–90% time.

Audit trail

Keep input identification and analysis settings visible.

Engineer sign-off

Provide space for review; draft is not certification.

Repeatability

Use the same process for the next weekly test.

EXAMPLE | NOT A PRODUCT ACCEPTANCE DECISION

Illustrative GEN1 vs GEN2 result

GEN1 / TEST A

10–90% response

1.688 s

GEN2 / TEST B

10–90% response

10.890 s

Selected events only; ~6.45× timing difference. Conditions and command/channel selection must be verified before drawing engineering conclusions.

Deployment: portable Windows engineering pilot

Builder machine

Package TTIC and dependencies on a Windows computer with Python.

Engineer machine

Run bundled application without separately installing Python.

Local browser

Launch the Streamlit interface on localhost.

Data handling

Process test files locally; no cloud upload in core workflow.

IT review

Corporate endpoint protection may still inspect/block executables.

Current status

Build kit prepared; Windows executable requires build and validation.

Pilot plan: validate before rollout

Week 1 • Setup

Build executable, launch on Windows, test representative uploads.

Week 1 • Accuracy

Cross-check raw statistics, rates, stability and event metrics.

Week 1 • Workflow

Run real comparisons and generate both report formats.

Feedback

Capture defects, confusing controls and missing needs.

Acceptance

Agree reproducibility, usability and critical-error criteria.

Decision

Fix findings, freeze pilot release, then assess wider deployment.

What TTIC is — and is not

TODAY

A local engineering analysis assistant

A repeatable comparison and reporting workflow

A pilot awaiting full numerical and user validation

NOT YET

A certified measurement system

An automatic product pass/fail authority

A validated enterprise deployment

The ask: one engineer, real data, structured feedback

Run a focused pilot with actual thermal test files. Verify the numbers, evaluate the workflow, and decide what is needed for a reliable wider release.

SUCCESS = TRUSTWORTHY RESULTS + FASTER REVIEW + ENGINEER CONFIDENCE